

Paired Patchwise Solver Comparisons

What is being compared

Each solver is evaluated on the same 100 patches, so the inferential unit is the patchwise paired difference. For metric M and solvers A and B , the tested quantity is

$$d_P = M_A(P) - M_B(P).$$

The descriptive mean/SD table is useful for orientation, but significance depends on the variance of these paired differences, not on the separate solver standard deviations. The inferential tables below compare the optimization solvers against the two interpolation baselines, IDW and ILDW.

The one-sided t tests for RMSE and signed Bias use

$$H_0 : \mathbb{E}[d] \geq 0, \quad H_1 : \mathbb{E}[d] < 0.$$

The paired-test tables use $\alpha = 0.01$. The one-sided p -value is the probability, under the boundary null $\mathbb{E}[d] = 0$, of observing a test statistic at least as favorable to solver A as the one observed. The p -value rule is to reject when the one-sided p -value is at most 0.01.

The 99% one-sided bound is a confidence-bound version of the same test. For RMSE and signed Bias, the table reports an upper bound for the true mean paired difference $\mathbb{E}[d]$; if that upper bound is below zero, then the whole 99% one-sided confidence region lies in the “solver A is smaller” direction. For Pearson correlation, the table reports a lower bound; if that lower bound is above zero, then the whole 99% one-sided confidence region lies in the “solver A has higher correlation” direction. These bound rules and the $p \leq 0.01$ rule give the same decisions because they are two equivalent forms of the same one-sided t test at the same 1% significance level.

d_3 bin (m)	mean pixels	IDW			ILDW			Nonlinear Opt.			VC Light Shrinkage			Homotopy Light Shrinkage		
		RMSE	Bias	Pearson r	RMSE	Bias	Pearson r	RMSE	Bias	Pearson r	RMSE	Bias	Pearson r	RMSE	Bias	Pearson r
<125	4830.6800	0.6525 ± 0.7805	-0.0194 ± 0.0668	0.9671 ± 0.0179	0.7722 ± 0.0105	-0.0014 ± 0.0635	0.9520 ± 0.0263	0.5106 ± 0.8270	0.0030 ± 0.0419	0.9810 ± 0.0203	0.3637 ± 0.4773	-1.153e-04 ± 0.0143	0.9888 ± 0.0086	0.3652 ± 0.4721	-4.286e-04 ± 0.0138	0.9888 ± 0.0086
(125,375]	1.900e+04	0.7297 ± 0.8627	-0.0158 ± 0.0646	0.9597 ± 0.0199	0.7269 ± 0.9108	-9.566e-04 ± 0.0230	0.9581 ± 0.0213	0.5603 ± 0.8503	-0.0100 ± 0.0218	0.9780 ± 0.0189	0.4137 ± 0.5104	-7.573e-04 ± 0.0159	0.9862 ± 0.0081	0.4136 ± 0.5098	-0.0012 ± 0.0157	0.9861 ± 0.0081
(375,750]	3.020e+04	0.9064 ± 1.0679	-0.0071 ± 0.1336	0.9411 ± 0.0286	0.8185 ± 0.9583	0.0052 ± 0.0921	0.9499 ± 0.0251	0.7005 ± 0.9790	-0.0444 ± 0.1012	0.9673 ± 0.0244	0.5336 ± 0.6475	0.0035 ± 0.0195	0.9780 ± 0.0121	0.5338 ± 0.6473	0.0029 ± 0.0195	0.9780 ± 0.0121
(750,1500]	5.844e+04	1.1086 ± 1.1395	-0.0149 ± 0.2070	0.9081 ± 0.0438	1.0099 ± 1.0550	-0.0026 ± 0.1643	0.9215 ± 0.0390	0.9170 ± 1.0660	-0.1366 ± 0.2144	0.9452 ± 0.0385	0.6764 ± 0.7163	0.0022 ± 0.0324	0.9634 ± 0.0210	0.6766 ± 0.7163	0.0017 ± 0.0324	0.9634 ± 0.0210
(1500,3125]	9.962e+04	1.4196 ± 1.3401	0.0035 ± 0.3578	0.8443 ± 0.0800	1.3469 ± 1.2881	0.0186 ± 0.3279	0.8591 ± 0.0740	1.3201 ± 1.3035	-0.2838 ± 0.4349	0.8943 ± 0.0743	0.9100 ± 0.8755	0.0112 ± 0.0835	0.9320 ± 0.0416	0.9100 ± 0.8754	0.0110 ± 0.0835	0.9320 ± 0.0416
(3125,6000]	5.213e+04	1.5957 ± 1.4288	0.1389 ± 0.6346	0.7761 ± 0.1257	1.5404 ± 1.3096	0.1459 ± 0.5838	0.7908 ± 0.1151	1.5502 ± 1.3604	0.2749 ± 0.6781	0.8399 ± 0.1101	1.0868 ± 0.9624	0.0688 ± 0.2232	0.8875 ± 0.0746	1.0868 ± 0.9625	0.0692 ± 0.2234	0.8875 ± 0.0746
(6000,9000]	4685.6200	1.4356 ± 1.3474	0.6299 ± 0.9829	0.6174 ± 0.3024	1.4165 ± 1.3035	0.6422 ± 0.9707	0.6237 ± 0.3039	1.2760 ± 1.2904	0.2375 ± 1.0179	0.6598 ± 0.2992	1.0292 ± 1.0765	0.3317 ± 0.7597	0.7463 ± 0.2450	1.0299 ± 1.0767	0.3349 ± 0.7591	0.7464 ± 0.2451
>9000	1090.6909	1.1345 ± 0.8346	0.7135 ± 0.8978	0.2872 ± 0.5728	1.1179 ± 0.7993	0.7203 ± 0.8713	0.2890 ± 0.5844	0.9822 ± 0.8306	0.4318 ± 0.9568	0.3258 ± 0.5681	0.7097 ± 0.5150	0.2394 ± 0.5701	0.2723 ± 0.6205	0.7088 ± 0.5133	0.2446 ± 0.5671	0.2730 ± 0.6196

Table 1: Distance-to-3rd-closest-link bin summaries. Each metric cell reports mean $\pm$ sample SD across patch-level bin metrics. RMSE, Bias, and Pearson r are recomputed from the underlying pixel values inside each d_3 distance bin for each patch and solver.

The sample size n can differ between tables, and sometimes even for the same distance bin, because each table counts only patch/bin entries where that metric is defined. A bin can contain a different number of valid pixels from patch to patch; Pearson correlation additionally requires at least two pixels with nonzero variation in both the ground truth and solver field. Therefore an RMSE or Bias entry can exist for a patch/bin where the corresponding Pearson entry is omitted.

Paired t analysis for RMSE

For RMSE, negative differences mean that solver A has lower RMSE than solver B on average. This is the natural one-sided direction for testing whether A improves on B .

Comparison	n	mean A	mean B	mean diff	SD diff	SE	t	one-sided p	99% upper bound	α	decision
Nonlinear Optimizer (300 iters, normal shrinkage) - IDW	100	1.1958	1.3067	-0.1108	0.1276	0.0128	-8.6854	3.814e-14	-0.0807	0.0100	reject
Nonlinear Optimizer (300 iters, normal shrinkage) - ILDW	100	1.1958	1.2390	-0.0432	0.0869	0.0087	-4.9659	1.428e-06	-0.0226	0.0100	reject
Virtual Convex Light Shrinkage (2000 iters) - IDW	100	0.8452	1.3067	-0.4615	0.4964	0.0496	-9.2953	1.166e-15	-0.3441	0.0100	reject
Virtual Convex Light Shrinkage (2000 iters) - ILDW	100	0.8452	1.2390	-0.3938	0.4402	0.0440	-8.9454	1.160e-14	-0.2897	0.0100	reject
Homotopy Light Shrinkage (2000 iters) - IDW	100	0.8452	1.3067	-0.4614	0.4965	0.0496	-9.2940	1.832e-15	-0.3440	0.0100	reject
Homotopy Light Shrinkage (2000 iters) - ILDW	100	0.8452	1.2390	-0.3937	0.4402	0.0440	-8.9438	1.021e-14	-0.2896	0.0100	reject

Table 2: Paired t comparisons for patchwise RMSE differences.

Paired t analysis for Bias

For Bias, the difference is the signed bias difference $Bias_A - Bias_B$. The one-sided lower-tail test asks whether solver A has a smaller signed bias than solver B . This is not the same as testing which solver is closer to zero bias; for that question, use paired differences of absolute bias instead.

Comparison	n	mean A	mean B	mean diff	SD diff	SE	t	one-sided p	99% upper bound	α	decision
Nonlinear Optimizer (300 iters, normal shrinkage) - IDW	100	-0.1863	0.0343	-0.2206	0.3083	0.0308	-7.1569	7.302e-11	-0.1477	0.0100	reject
Nonlinear Optimizer (300 iters, normal shrinkage) - ILDW	100	-0.1863	0.0468	-0.2331	0.2912	0.0291	-8.0063	1.163e-12	-0.1643	0.0100	reject
Virtual Convex Light Shrinkage (2000 iters) - IDW	100	0.0233	0.0343	-0.0110	0.2445	0.0245	-0.4513	0.3264	0.0468	0.0100	do not reject
Virtual Convex Light Shrinkage (2000 iters) - ILDW	100	0.0233	0.0468	-0.0235	0.2083	0.0208	-1.1296	0.1307	0.0257	0.0100	do not reject
Homotopy Light Shrinkage (2000 iters) - IDW	100	0.0231	0.0343	-0.0112	0.2445	0.0245	-0.4581	0.3240	0.0466	0.0100	do not reject
Homotopy Light Shrinkage (2000 iters) - ILDW	100	0.0231	0.0468	-0.0237	0.2084	0.0208	-1.1375	0.1290	0.0256	0.0100	do not reject

Table 3: Paired t comparisons for patchwise signed-bias differences.

Paired Fisher- z analysis for correlation

For Pearson correlation, first apply the Fisher transform

$$z = \frac{1}{2} \log \frac{1+r}{1-r},$$

then compare the paired z values across patches. For correlation, larger is better, so the Fisher- z table uses the one-sided direction $H_0 : \mathbb{E}[d] \leq 0$ against $H_1 : \mathbb{E}[d] > 0$.

Comparison	n	mean A	mean B	mean diff	SD diff	SE	t	one-sided p	99% lower bound	α	decision
Nonlinear Optimizer (300 iters, normal shrinkage) - IDW	100	1.5563	1.3678	0.1885	0.1243	0.0124	15.1613	2.442e-15	0.1591	0.0100	reject
Nonlinear Optimizer (300 iters, normal shrinkage) - ILDW	100	1.5563	1.4161	0.1402	0.1147	0.0115	12.2162	1.554e-15	0.1130	0.0100	reject
Virtual Convex Light Shrinkage (2000 iters) - IDW	100	1.8171	1.3678	0.4493	0.1293	0.0129	34.7495	1.998e-15	0.4187	0.0100	reject
Virtual Convex Light Shrinkage (2000 iters) - ILDW	100	1.8171	1.4161	0.4009	0.1193	0.0119	33.6160	1.110e-15	0.3727	0.0100	reject
Homotopy Light Shrinkage (2000 iters) - IDW	100	1.8171	1.3678	0.4494	0.1291	0.0129	34.8183	2.220e-15	0.4188	0.0100	reject
Homotopy Light Shrinkage (2000 iters) - ILDW	100	1.8171	1.4161	0.4010	0.1191	0.0119	33.6749	1.554e-15	0.3728	0.0100	reject

Table 4: Paired t comparisons for Fisher-transformed patchwise Pearson correlations. Positive differences mean solver A has higher transformed correlation than solver B .

Comparison	nonzero n	A higher	B higher	one-sided p
Nonlinear Optimizer (300 iters, normal shrinkage) - IDW	100	98	2	3.985e-27
Nonlinear Optimizer (300 iters, normal shrinkage) - ILDW	100	100	0	7.889e-31
Virtual Convex Light Shrinkage (2000 iters) - IDW	100	100	0	7.889e-31
Virtual Convex Light Shrinkage (2000 iters) - ILDW	100	100	0	7.889e-31
Homotopy Light Shrinkage (2000 iters) - IDW	100	100	0	7.889e-31
Homotopy Light Shrinkage (2000 iters) - ILDW	100	100	0	7.889e-31

Table 5: One-sided sign test for patchwise Pearson correlation differences. The alternative is that solver A has higher Pearson correlation than solver B on more than half of the non-tied patches.